Unit – I

INTRODUCTION TO MODEL BUILDING

An Introduction to Modeling:

Operations research (often referred to as management science) is simply a scientific approach to decision making that seeks to best design and operate a system, usually under conditions requiring the allocation of scarce resources.

By a system, we mean an organization of interdependent components that work together to accomplish the goal of the system. For example, Ford Motor Company is a system whose goal consists of maximizing the profit that can be earned by producing quality vehicles. The term operations research was coined during World War II when British military leaders asked scientists and engineers to analyze several military problems such as the deployment of radar and the management of convoy, bombing, antisubmarine, and mining operations.

The scientific approach to decision making usually involves the use of one or more mathematical models. A mathematical model is a mathematical representation of an actual situation that may be used to make better decisions or simply to understand the actual situation better. The following example should clarify many of the key terms used to describe mathematical models.

Prescriptive or Optimization Models:

Most of the models discussed in this book will be prescriptive or optimization models. A prescriptive model "prescribes" behavior for an organization that will enable it to best meet its goal(s). The components of a prescriptive model include

i. Decision variables
ii. Objective function(s)
iii. Constraints

In short, an optimization model seeks to find values of the decision variables that optimize (maximize or minimize) an objective function among the set of all values for the decision variables that satisfy the given constraints.

The Seven-Step Model-Building Process:

When operations research is used to solve an organization's problem, the following seven step model-building procedure should be followed:

Step 1: Formulate the Problem The operations researcher first defines the organization's problem. Defining the problem includes specifying the organization's objectives and the parts of the organization that must be studied before the problem can be solved.

Step 2: Observe the System Next, the operations researcher collects data to estimate the value of parameters that affect the organization's problem. These estimates are used to develop (in step 3) and evaluate (in step 4) a mathematical model of the organization's problem.

Step 3: Formulate a Mathematical Model of the Problem In this step, the operations researcher develops a mathematical model of the problem. In this book, we will describe

many mathematical techniques that can be used to model systems.

Step 4: Verify the Model and Use the Model for Prediction The operations researcher now tries to determine if the mathematical model developed in step 3 is an accurate representation of reality.

Step 5: Select a Suitable Alternative Given a model and a set of alternatives, the operations researcher now chooses the alternative that best meets the organization's objectives.

Step 6: Present the Results and Conclusion of the Study to the Organization In this step, the operations researcher presents the model and recommendation from step 5 to the decision making individual or group. In some situations, one might present several alternatives and let the organization choose the one that best meets its needs. After presenting the results of the operations research study, the analyst may find that the organization does not approve of the recommendation. This may result from incorrect definition of the organization's problems or from failure to involve the decision maker from the start of the project.

Step 7: Implement and Evaluate Recommendations If the organization has accepted the study, then the analyst aids in implementing the recommendations. The system must be constantly monitored (and updated dynamically as the environment changes) to ensure that the recommendations enable the organization to meet its objectives.

INTRODUCTION TO OPERATIONS RESEARCH

Introduction:

Operations Research is the science of rational decision-making and the study, design and integration of complex situations and systems with the goal of predicting system behavior and improving or optimizing system performance.

Operations Research tools are not from any one discipline. Operations Research takes tools from different discipline such as mathematics, statistics, economics, psychology, engineering etc. and combines these tools to make a new set of knowledge for decision making. Today,
O.R. became a professional discipline which deals with the application of scientific methods for making decision, and especially to the allocation of scarce resources. The main purpose of
O.R. is to provide a rational basis for decisions making in the absence of complete information, because the systems composed of human, machine, and procedures may do not have complete information.

Definitions:

The definitions stressed by various experts and Societies on the subject together enable us to know what Operations Research is, and what it does. They are as follows:

i. According to Operations Research Society of America (ORSA): Operations Research is concerned with scientifically deciding how to best design and operate man-machine systems usually under conditions requiring the allocation of scarce resources.
ii. Operations Research is the art of winning wars without actually fighting. - Aurther Clarke
iii. Operations Research is the "art of giving bad answers to problems which otherwise have worse answers". - T.L.Saaty
iv. Operations Research is applied decision theory. It uses any scientific, mathematical or logical means to attempt to cope with the problems that confront the executive, when he tries to achieve a thorough-going rationality in dealing with his decision problems. - D.W.
Miller and M.K.Starr
v. Operations Research is a "scientific approach to problems solving for executive management". - H.M. Wagner
vi. Operations Research is the "application of scientific methods, techniques and tools to problems involving the operations of a system so as to provide those in control of the system with optimum solution to the problem". - Churchman, Ackoff and Arnoff
vii. Operations Research is "scientific methodology-analytical, experimental, quantitative-which by assessing the overall implication of various alternative courses of action in a management system, provides an improved basis for management decisions". – Pocock

Phases and Processes of Operations Research:

i. **Formulate the problem:** This is the most important process, it is generally lengthy and time consuming. The activities that constitute this step are visits, observations, research, etc. With the help of such activities, the O.R. scientist gets sufficient information and support to proceed and is better prepared to formulate the problem. This process starts with understanding of the organizational climate, its objectives and expectations. Further,the alternative courses of action are discovered in this step.

ii. **Develop a model:** Once a problem is formulated, the next step is to express the problem into a mathematical model that represents systems, processes or environment in the form of equations, relationships or formulas. We have to identify both the static and dynamic structural elements, and device mathematical formulas to represent the interrelationships among elements. The proposed model may be field tested and modified in order to work under stated environmental constraints. A model may also be modified if the management is not satisfied with the answer that it gives.

iii. **Select appropriate data input:** Garbage in and garbage out is a famous saying. No model will work appropriately if data input is not appropriate. The purpose of this step is to have sufficient input to operate and test the model.

iv. **Solution of the model:** After selecting the appropriate data input, the next step is to find a solution. If the model is not behaving properly, then updating and modification is considered at this stage.

v. **Validation of the model:** A model is said to be valid if it can provide a reliable prediction of the system's performance. A model must be applicable for a longer time and can be updated from time to time taking into consideration the past, present and future aspects of the problem.

vi. **Implement the solution:** The implementation of the solution involves so many behavioral issues and the implementing authority is responsible for resolving these issues. The gap between one who provides a solution and one who wishes to use it should be eliminated. To achieve this, O.R. scientist as well as management should play a positive role. A properly implemented solution obtained through O.R. techniquesresults in improved working and wins the management support.

Applications of Operations Research:

i. Finance, Budgeting and Investments
   a. Credit policy analysis.
   b. Cash flow analysis.
   c. Dividend policies.
   d. Investment portfolios.

ii. Marketing
   a. Product selection, timing, etc.
   b. Advertising media, budget allocation.
   c. Number of salesman required.
   d. Selection of product mix.

iii. Purchasing, Procurement and Exploration
   a. Optimal buying and reordering.
   b. Replacement policies

iv. Production Management
   a. Location and size of warehouses, factories, retail outlets, etc.
   b. Distribution policy.
   c. Loading and unloading facilities for trucks, etc.
   d. Production scheduling.
   e. Optimum product mix.
   f. Project scheduling and allocation of resources.

v. Personnel Management
   a. Selection of suitable personnel.
   b. Recruitment of employees.
   c. Assignment of jobs.

vi. Research and Development
   a. Project selection.
   b. Control of R&D projects.
   c. Reliability and alternative design.

Journals of Operations Research: There are twelve scholarly journals for operationsresearch. They are:

i. Decision Analysis.
ii. Information Systems Research.
iii. INFORMS Journal on Computing.
iv. INFORMS Transactions on Education.
v. Interfaces.
vi. Management Science.
vii. Manufacturing & Service Operations Management
viii. Marketing Science.
ix. Mathematics of Operations Research.
x. Operations Research.
xi. Organization Science.
xii. Transportation Science.

The **other journals** are:

i. INFOR Journal.
ii. Journal of Defense Modeling and Simulation (JDMS).
iii. Journal of the Operational Research Society (JORS).
iv. Journal of Simulation (JOS).
v. Mathematical Methods of Operations Research (MMOR).
vi. Military Operations Research (MOR).
vii. Opsearch.
viii. OR Insight.
ix. TOP: official journal of the Spanish Society of Statistics and Operations Research.

LINEAR PROGRAMMING

Introduction:

Linear Programming (LP) is a versatile technique for assigning a fixed amount of resources among competing factors, in such a way that some objective is optimized and other defined conditions are also satisfied. In other words, linear programming is a mathematical technique for determining the optimal allocation of resources and obtaining a particular objective when there are alternative uses of the resources. The objective may be cost minimization or inversely profit maximization.

Basic Terminology:

The present section serves the purpose of building your vocabulary of the terms frequently employed in the description of Linear Programming Models.

1. ***Linear Function:*** A linear function contains terms each of which is composed of only a single, continuous variable raised to (and only to) the power of 1.
2. ***Objective Function:*** It is a linear function of the decision variables expressing the objective of the decision-maker. The most typical forms of objective functions are: maximize f(x) or minimize f(x).
3. ***Decision Variables:*** These are economic or physical quantities whose numerical values indicate the solution of the linear programming problem. These variables are under the control of the decision-maker and could have an impact on the solution to the problem under consideration. The relationships among these variables should be linear.
4. ***Constraints:*** These are linear equations arising out of practical limitations. The mathematical forms of the constraints are: $f(x) \geq b$ or $f(x) \leq b$ or $f(x) = b$
5. ***Feasible Solution:*** Any non-negative solution which satisfies all the constraints is known as a feasible solution. The region comprising all feasible solutions is referred to as feasible region.
6. ***Optimal Solution:*** The solution where the objective function is maximized or minimized is known as optimal solution.

Mathematical Formulation: Consider the followingOptimize (Maximize or Minimize)

$$Z = c_1x_1 + c_2x_2 + c_3x_3 + \ldots\ldots\ldots + c_nx_n$$

Subject to

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots\ldots + a_{1n}x_n \ (\leq,=,\geq)\ b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots\ldots + a_{2n}x_n \ (\leq,=,\geq)\ b_2$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$

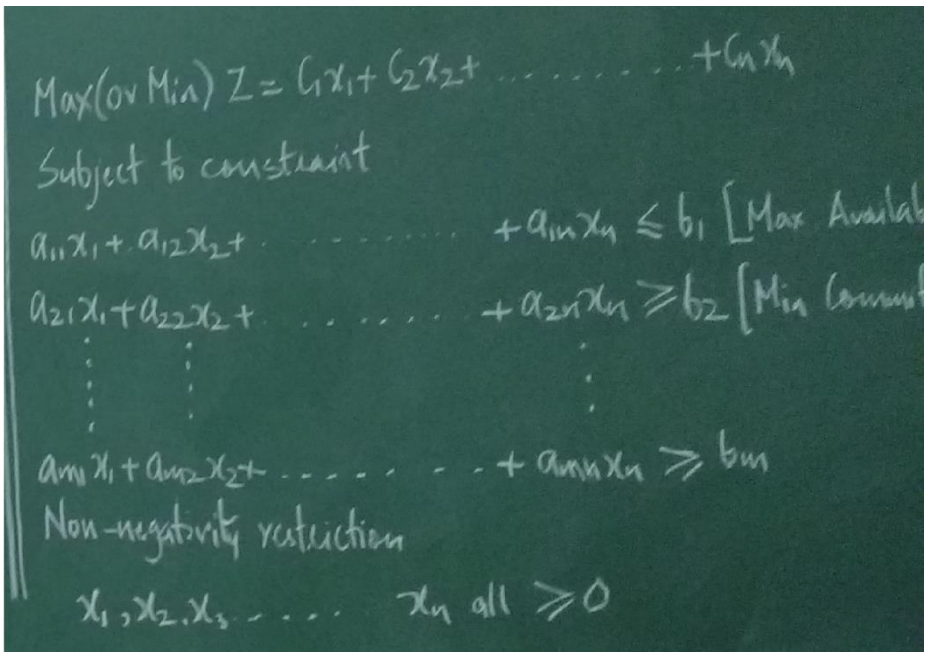
$$a_{m1}x_1 + a_{m2}x_2 + a_{m3}x_3 + \ldots + a_{mn}x_n \ (\leq,=,\geq)\ b_m$$

$$x_1 , x_2 , \ldots , x_n \geq 0$$

Where c_j's, a_{ij}'s are constants and x_j's are decision variables. The expression ($\leq$, =, $\geq$) meanthat each constraint may take only one of the three possible forms:

i. Less than or equal to ($\leq$)
ii. Equal to (=)
iii. Greater than or equal to ($\geq$)

The expression $x_i \geq 0$ means that x_i's must be non - negative.



Assumptions:

i. ***Additivity:*** The total amounts of each input and the corresponding profit are the sums of the inputs and profit for each individual process. For example, one unit of Product 1 and one unit of Product 2 require 3 (2 + 1) hours of processing in machine 1, 7 (3 + 4) hours of processing in machine 2, 11 (4 + 7) hours of processing in machine 3, and yield Rs.110 (50 + 60) profit.

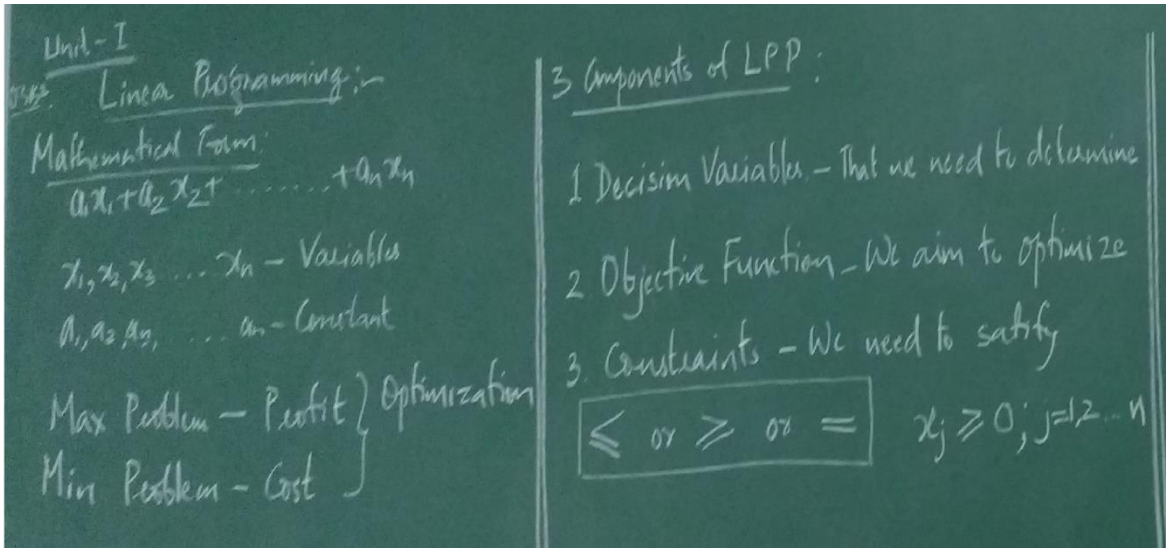
ii. ***Divisibility:*** The activity levels are allowed to assume fractional values as well as integer values. For example, we admit the possibility of x_1 = 509/3 or 169.67, x_2 = 509/4 or 127.25.

iii. ***Proportionality:*** The total amounts of each input and the associated profit are directly proportional to the level of output.

iv. ***Certainty:*** The certainty assumption requires that the parameters cj, bi, and a_{ij} be known constants. The optimum solution derived is predicated on perfect knowledge of all the parameter values. Since all exogenous factors are assumed to be known and fixed, LP models are sometimes called non-stochastic as contrasted with models explicitly dealing with stochastic factors. This assumption gives rise to the term "deterministic" analysis.

Definitions:

1. ***Solution*** - Any specification of the values for decision variable among (x_1 , x_2..... x_n) is called a solution.
2. ***Feasible solution*** is a solution for which all constraints are satisfied.
3. ***Infeasible solution*** is a solution for which atleast one constraint is not satisfied.
4. ***Feasible region*** is a collection of all feasible solutions.
5. ***Optimal solution*** is a feasible solution that has the most favorable value of the objective function.
6. ***Most favorable value*** is the largest value if the objective function is to be maximized, whereas it is the smallest value if the objective function is to be minimized.
7. ***Multiple optimal solution*** - More than one solution with the same optimal value of the objective function.
8. ***Unbounded solution*** - If the value of the objective function can be increased ordecreased indefinitely such solutions are called unbounded solution.
9. ***Feasible region*** - The region containing all the solutions of an inequality
10. ***Corner point feasible solution*** is a solution that lies at the corner of the feasible region.



Problems on Production:

1. A firm manufacture two type of products A & B and sells them at a profit of Rs. 2 on type A & Rs. 3 on type B. Each product is processed on two machines G & H. Type A requires 1 minute of processing time on G & 1 minute on H. The machine G is available for not more than 6 hours 40 minutes while machine H is available for 10 hours during any working days. Formulate the problem as a Linear Programming Problem.

Solution:

Let x_1 be the number of products of type Ax_2 be the number of products of type B

Since the profit on type A is Rs.2 per product, $2x_1$ will be the profit on selling x_1 units of typeA.

Similarly, $3x_2$ will be the profit on selling x_2 units of type B.

Therefore, total profit on selling x_1 units of A and x_2 units of type B is given byMaximize $Z = 2x_1 + 3x_2$ (objective function)

Since machine G takes 1 minute time on type A and 1 minute time on type B, the total number of minutes required on machine G is given by $x_1 + x_2$.

Similarly, the total number of minutes required on machine H is given by $2x_1 + 3x_2$ .But, machine G is not available for more than 6 hours 40 minutes (400 minutes).

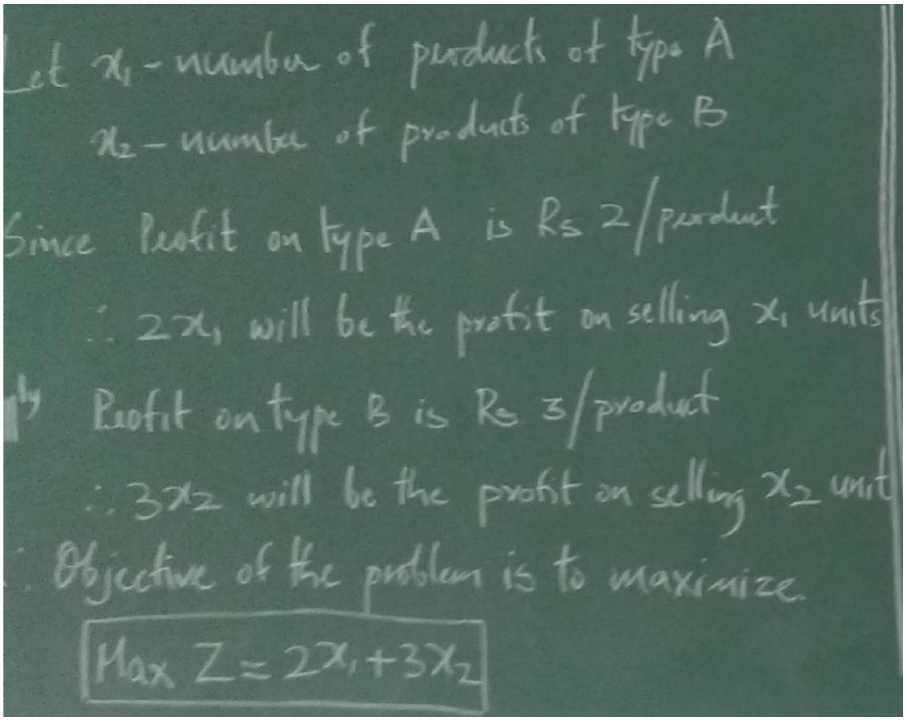
Therefore, $x_1 + x_2 \leq 400$ (first constraint)

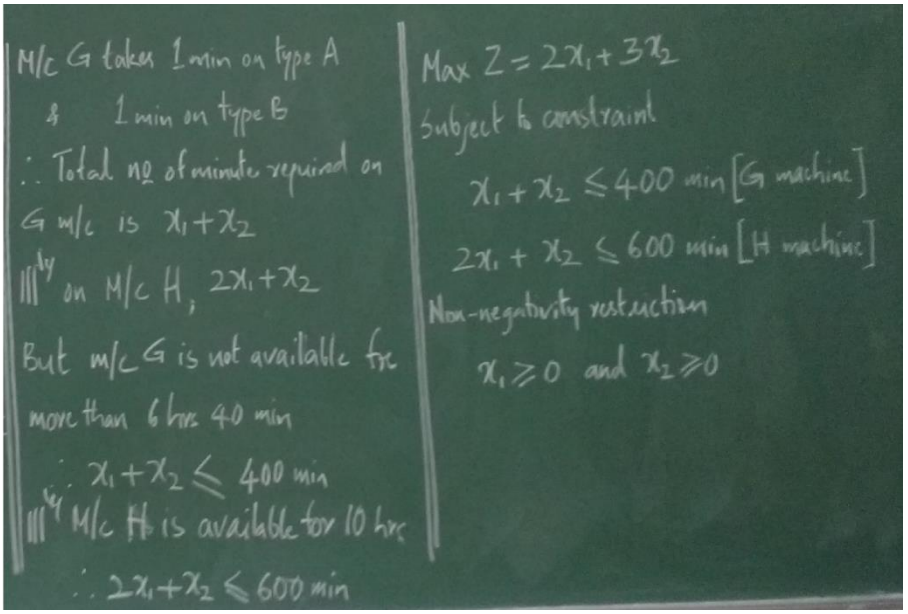
Also, the machine H is available for 10 hours (600 minutes) only,Therefore, $2x_1 + 3x_2 \leq 600$ (second constraint)

Since it is not possible to produce negative quantities$x_1 \geq 0$ and $x_2 \geq 0$ (non-negative restrictions)

Hence

Maximize $Z = 2x_1 + 3x_2$
Subject to restrictions $x_1 + x_2 \leq 400$
$2x_1 + 3x_2 \leq 600$
and non-negativity constraints$x_1 \geq 0$, $x_2 \geq 0$





2. A company owns 2 oil mills A and B which have different production capacities for low, high and medium grade oil. The company enters into a contract to supply oil to a firm every week with 12, 8, 24 barrels of each grade respectively. It costs the company Rs 1000 and Rs 800 per day to run the mills A and B. On a day A produces 6, 2, 4 barrels of each grade and B produces 2, 2, 12 barrels of each grade. Formulate an LPP to determine number of days per week each mill will be operated in order to meet the contract economically.

3. A company sells two different products A and B, making a profit of Rs.40 and Rs.30 per unit on them respectively. They are produced in a common production process and are sold in two different markets. The production process has a total capacity of 30,000 man-hours. It takes three hours to produce a unit of A and one hour to produce a unit of B. The market has been surveyed and company officials feel that the maximum number of units of A that can be sold is 8,000 units and that of B is 12,000 units. Subject to these limitations, products can be sold

in any combination. Formulate this problem as a linear programming model to maximize the profit.

4. The manager of an oil refinery must decide on the optimal mix of two possible blending processes of which the input and output per production run are given as follows:

Process (units)	Input (units)		Output	
	Crude A	Crude B	Gasoline X	Gasoline Y
1	5	3	5	8
2	4	5	4	4

The maximum amount available of crude A and B are 200 units and 150 units respectively. Market requirements show that at least 100 units of gasoline X and 80 units of gasoline Y must be produced. The profit per production run from process 1 and process 2 are Rs.300 and Rs.400 respectively. Formulate this problem as a linear programming model to maximize the profit.

5. A firm places an order for a particular product at the beginning of each month and the product is received at the end of the month. The firm sells during the month from the stocks and it can sell any quantity.
The prices at which the firm buys and sells vary every month. The following table shows the projected buying and selling prices for the next four months:

Month	Selling price(Rs) (during the month)	Purchase price (Rs) (beginning of the month)
April	-	75
May	90	75
June	60	60
July	75	-

The firm has no stocks on hand as on April 1, and does not wish to have any stocks at the end of July. The firm has a warehouse of limited size, which can hold a maximum of 150 units of the product.
The problem is to determine the number of units to buy and sell each month to maximize the profits from its operations. Formulate this problem as a linear programming problem to maximize profits from its operations.

6. A manufacturer produces three models (I, II and III) of a certain product. He uses two types of raw material (A and B) of which 4000 and 6000 units respectively are available. The raw material requirements per unit of the three models are as follows:

Raw material	Requirement per unit of given model		
	I	II	III
A	2	3	5
B	4	2	7

The labour time of each unit of model I is twice that of model II and three times that of the model III. The entire labour force of the factory can produce the equivalent of 2500 units of model I. A market survey indicates that the minimum demands of the three models are: 500,500 and 375 units respectively. However, the ratios of the number of units produced must be equal to 3:2:5. Assume that the profit per unit of models I,II and III are Rs60,40 and 100 respectively. Formulate this problem as an LP model in order to determine the number of units of each product which will maximize profit

7. Consider a small plant which makes two types of automobile parts, say A and B. it buys castings that are machined, bored and polished. The capacity of machining is 25 per hour for A and 24 per hour for B, capacity of boring is 28 per hour for A and 35 per hour for B, and the capacity of polishing is 35per hour for A and 25 per hour for B. Castings for part A cost Rs 2 and sell for Rs 5 each and those for part B cost Rs 3 and sell for Rs 6 each. The three machines have running costs of Rs 20, Rs 14 and Rs 17.50 per hour. Assuming that any combination of parts A and B can be sold, formulate this problem as an LP model to determine the product mix which maximizes profit.

8. On October 1, a company received a contract to supply 6,000 units of a specialized product. The terms of contract require that 1,000 units be shipped in October; 3,000 Units in November and 2,000 units in December. The company can manufacture 1,500 units per month on regular time and 750 units per month on overtime. The manufacturing cost per item produced during regular time is Rs 3 and the cost per item produced during overtime is Rs 45. The monthly storage cost is Re 1. Formulate this problem as an LP model so as to minimize the total cos.

9. A wine maker has a stock of three different wines with the following characteristics:

Wine	Proofs	Acids %	Specific gravity	Stock (Gallons)
A	27	0.32	1.70	20
B	33	0.20	1.08	34
C	32	0.30	1.04	22

A good dry table wine should be between 30 and 31 degree proof. It should contain at least 0.25% acid and should have a specific gravity of at least 1.06. The wine maker wishes to blend the three types of wine to produce as large as a quantity as possible of a satisfactory dry table wine. However, his stock of wine A must be completely used in the blend because further storage would cause it to deteriorate. What quantities of wines B and C should be used? Formulate the problem as an LP model.

10. The XYZ Company assembles and markets two types of transistor radios, A and B. Presently 200 radios of each type are manufactured per week. You are advised to formulate the production schedule which will maximize the profits in the light of the following information.

Type	Total component costper radio(Rs)	Man-hours of assembly time per radio	Average man-minutes of inspection and correction time per radio	Selling price per radio(Rs)
A	200	12	10	400
B	160	6	35	280

The company employs 100 assembles who are paid Rs 10 per hour actually worked and who will work up to a maximum of 48hours per week. The inspectors, who are presently four, have agreed to a plan whereby they average 40hours of work per week each. However, the four inspectors have certain other administrative duties which have been found to take up an average of 8hours per week between them. The inspectors are each paid a fixed wages of Rs 800 per week.

Each radio of either type requires one speaker, the type being the same for each radio. However, the company can obtain a maximum supply of 600 in any one week. Their cost has been included in the components cost given for each radio in the table above. Only speakers actually used need be paid for. The only other cost incurred by the company is fixed overheads of Rs 20,000 per week.

11. ABC Foods Company is developing a low-calorie high-protein diet supplement called Hi-Pro.

The specification for Hi-Pro has been established by a panel of medical experts. These specification along with the calorie, protein and vitamin content of three basic foods, are given in the following table:

Nutritional elements	Units of nutritional elements(per 100g serving of basic foods)			Hi-Pro specification
	Basic foods			
	1	2	3	
Calories	350	250	200	300
Proteins	250	300	150	200
Vitamin A	100	150	75	100
Vitamin C	75	125	150	100
Cost per serving(Rs)	1.50	2.00	1.20	

What quantities of foods 1, 2, and 3 should be used? Formulate this problem as an LPmodel to minimize cost.

12. Omega leather goods company manufactures two types of leather soccer balls X and Y. Each type of balls requires work by both types of employees: semi-skilled and skilled. Basically, the semi-skilled employees employ machines in the manufacture of the balls, while the skilled employees hand sew balls. The available time (per week) for each type of employee and the time requirement for each type of ball are given below.

Type of employee	Manufacturing time requirement (hr)		Time available (hr/week)
	Ball X	Ball Y	
Semi-skilled	2	3	80
Skilled	4	6	150

The cost of an hour of semi-skilled labour is Rs 5.50 and that of an hour of skilled labour is Rs 8.50. To meet weekly demand requirements, at least 15 balls of type X and at least 10 balls of type Y must be manufactured. Formulate this problem as an LP model so as to minimize cost of production.

13. A pharmaceuticals company has developed a new pill to be taken by smokers that will nauseate them if they smoke. This new pill is a combination of four ingredients that are costly and in limited supply. The supply availabilities and costs are as follows:

Ingredient	Supply availability(Kg)	Cost (Rs/Kg)
1	22	28
2	18	25
3	20	52
4	24	26

Blending requirements for this new pill are as follows:
Ingredient 1 must be at least 45% of the total, but cannot exceed 60% of the total.
Ingredient 2 and 3 must each comprise at least 10% of the mixture, but their combined percentage cannot exceed 25% of the total.
Ingredient 4 must not be more than 50% of the total.
Additionally, at least 25kg of the pill must be produced. Formulate this problem as LP model.

14. The plastic manufacture has 1200 boxes of transparent wrap in stock at one factory and other boxes at its second factory. The manufacturer has orders for this product from three different retailers, in quantities of 1000,700 and 500boxes, respectively. The unit shipping costs (in rupees per box) from the factories to the retailers are as follows:

	Retailer I	Retailer II	Retailer III

Factory A	14	13	11
Factory B	13	13	12

Determine a minimum cost shipping schedule for satisfying all demands from current inventory. Formulate this problem as a linear programming model.

15. A paint manufacturing company manufactures paints at two plants. Firm orders have been received from three large contractors. The firm has determined that the following shipping cost data is appropriate for these contractors with respect to its twoplants.

Contractor	Order size (gallon)	Shipping cost/gallon(Rs)	
		From plant 1	From plant 2
A	750	1.80	2.00
B	1500	2.60	2.20
C	1500	2.10	2.25

Each gallon of paint must be blended and tinted. The company's costs with respect tothese two operations at each of the two plants are as follows:

Plant/Operation		Hours required per gallon	Cost/hours (Rs)	Hours available
Blending	Plant1	0.10	3.80	300
Tinting		0.25	3.20	360
Blending	Plant 2	0.15	4.00	600
Tinting		0.20	3.10	720

Formulate this problem as a linear programming model.

16. A company produces two types of sauce: A and B. These sauces are both made blending two ingredients X and Y. A certain level of flexibility is permitted in the formulae of these products. Indeed, the restrictions are that (i) B must contain no morethan 75% of X, and (ii) A must contain no less than 25% of A, and no less than 50% of B. Up to 400kg of X and 300kg of Y could be purchased. The company can sell as much of these sauces at it produces at a price of Rs 18 for A and Rs 17 for B. The X and Y cost Rs 1.60 and 2.05 per kg respectively. The company wishes to maximize its net revenue from the sale of these sauces. Formulate this problem at LP model.

17. Vitamins A and B are found in foods F_1 and F_2. One unit of food F_1 contains three units of vitamin A and four units of vitamin B. One unit of food F_2 contains six units of vitamin A and three units of vitamin B. One unit of food F_1 and F_2 cost Rs.4 and 5 respectively. The minimum daily requirement (for a person) of vitamin A and B is 80 and 100 units respectively. Assuming that anything in excess of daily minimum requirement of A and B is not harmful, find out the optimum mixture of food F_1 andF_2at the minimum cost which meets the daily minimum requirement of vitamins A and
B. Formulate this problem as a linear programming problem.

18. The Omega Data Processing Company performs three types of activity: payrolls, account receivables, and inventories. The profit and time requirements for keypunch computation and office printing for a 'standard job' are shown in the following table:

Job	Profit/Standard job (Rs)	Time requirement (Min.)		
		Keypunch	Computation	Print
Payroll	275	1,200	20	100

A/C Receivable	125	1,400	15	60
Inventory	225	800	35	80

Omega guarantees overnight completion of the job. Any job scheduled during the day can be completed during the day or night. Any job scheduled during the night, however, must be completed during the night. The capacity for both day and night is shown in the following table:

Capacity (Min)	Keypunch	Computation	Print
Day	4,200	150	400
Night	9,200	250	650

Formulate the linear programming problem in order to determine the 'mixture' of standard jobs that should be accepted during the day and night.

19. PQR Coffee Company mixes South Indian, Assamese and imported coffees to make two brands of coffee: plains X and plains XX. The characteristics used in blending the coffee include strength, acidity, and caffeine. The test results of the available supplies of South Indian, Assamese and imported coffee are shown in the following table:

Price per kg(Rs)	Price per kg(Rs)	Strength Index	Acidity Index	Percent caffeine	Supply available (kg)
South Indian	30	6	4.0	2.0	40000
Assamese	40	8	3.0	2.5	20,000
Imported	35	5	3.5	1.5	15,000

The requirements for plains X and plains XX coffees are given in the following table:

Plains coffee	Price per kg (Rs)	Minimum Strength caffeine	Maximum acidity (kg)	Maximum Per cent	Quantity demanded
X	45	6.5	3.8	2.2	35,000
XX	55	6.0	3.5	2.0	25,000

Assume that 35,000kg of plains X and 25,000kg of plains XX are to be sold.Formulate this problem as an LP model to maximize sales.

20. A manufactures of metal office equipment makes desks, chairs, cabinets and book cases. The work is carried out in the three major departments: Metal stamping, Assembly and Finishing. Exhibits, A, B and C give requisite data of the problem.

Exhibit A

Department	Time required in hours per unit of product				Hours available per week
	Products				
	Desk	Chair	Cabinet	Bookcase	
Stamping	4	2	3	3	800
Assembly	10	6	8	7	1,200
Finishing	10	8	8	8	800

Exhibit B

Department	Cost (Rs) of operation per unit of product			
	Products			
	Desk	Chair	Cabinet	Bookcase
Stamping	15	8	12	12
Assembly	30	18	24	21
Finishing	35	28	25	21

Exhibit C: selling price (Rs) per unit product

Desk: 175 Chair: 95
Cabinet: 145 Book case: 130

In order to maximize weekly profits, what should be the products program? Assuming that the items produced can be sold, which department needs to be expanded for increasing profits?

Problems on Marketing:

21. Suppose a media specialist has to decide on the allocation of advertising in three media vehicles. Let Xi be the number of message carried in the media=1, 2, 3. The unit costs of a message in the three media are Rs 1,000; Rs 750 and Rs 500. The total budget available is Rs 20,000 for the campaign period of a year. The first media is a monthly magazine and it is desired to advertise not more than one insertion in one issue. At least six message should appear in the second media. The number of message in the third media should strictly lie between 4 and 8. The expected effective audience for unit message in the media vehicles is shown below:

Vehicle	Expected effective audience
1	80,000
2	60,000
3	45,000

Formulate a linear programming model to determine the optimum allocation that would maximize total effective audience.

22. An advertising company is planning an advertisement campaign for a new product recently introduced in the market. It is decided to insert advertisements in three leading magazines. The company has made a careful analysis of three available media, and has compiled the following set of relevant data:

Data Item	Magazine		
	1	2	3
Reader characteristics			
a) Age:25-35 yrs	70%	80%	40%
b) Education level: Graduation and above	80%	60%	50%
c) Income: Rs 1,500 and above	60%	70%	40%
Minimum number of advertisements	20	10	5
Maximum number of advertisements	50	40	30
Cost per advertisement(Rs)	1,000	700	500
Readership	3,20,000	5,00,000	2,00,000

Additionally, the client has specified that relative importance of the reader characteristics should be weighed as follows:

Reader characteristics	Weightage
Age:25-35 yrs	0.4
Graduate and above	0.4
Income ≥ Rs 1,500	0.2

The client has Rs 50,000 to spend at this point in time. Formulate this problem as anLP model to maximize the effective exposure level.

23. The owner of Metro Sports wishes to determine how many advertisements to place in the selected three monthly magazines A, B, C. His objective is to advertise in such a way that total exposure to principal buyers to expensive sports goods is maximized. The percentage of readers for each magazine is known. Exposure to any particular magazine is the number

of advertisements placed multiplied by the number of principal buyers. The following data may be used:

	Magazines		
	A	B	C
Readers	1 lakh	0.6 lakh	0.4 lakh
Principal buyers	10%	15%	7%
Cost per advertisements(Rs)	5,000	4,500	4,250

The budget amount is at the most Rs 1 lakh for the advertisements. The owner has already decided that magazine A should have no more than six advertisements and that B and C each at least two advertisements. Formulate this problem as an LP model to determine the number of advertisements to be placed in each magazine.

24. A marketing company dealing with laminated sheets 'Gloss' in the western zone covering Maharashtra, Gujarat and Madhya Pradesh is considering to lunch an advertisement campaign within their budget of Rs 2.5 lakh. On the basis of advertisement testing of the previous year, the company's research department has found that magazines and films are the ideal media for advertisements of laminate sheets. The company is not in a position to use the audiovisual medium due to limitation of funds. The magazines enjoying good recall in last year's campaign are Stardust, Filmfare, Reader's Digest and Madhuri. This is attributed to the effective visual impact made by the good reproduction of the advertisements both in colour andblack and white.

The characteristic of target audience for 'Gloss' and weightage for each are as follows:

	Characteristics	Weightage (%)
Age	15-34	20
Monthly income	Over Rs1,000	70
Education	Above S.S.C	10

The audience characteristics for the four magazines selected are given below:

Characteristics	Stardust (%)	Film fare (%)	Reader's Digest (%)	Madhuri (%)
Age:15-34 yrs	75	45	56	80
Monthly income: Over Rs 1,000	52	43	47	25
Education: Above S.S.C	83	53	72	34

The efficacy index for a black and white advertisement may be taken as 0.15 and that for a colour advertisement 0.20. The cost per insertion of a black and white and a colour advertisement and the readership for the four magazines are as follows:

Magazines	Cost(Rs) per insertion		Readership (in '000)
	Black & White	Colour	
Stardust (Monthly)	4,500	8,400	189
Film fare(Fortnightly)	4,200	8,400	256
Reader's Digest (Monthly)	6,400	9,600	136
Madhuri (Fortnightly)	3,300	6,600	205

It has also been found that at least three insertions are necessary in Stardust and Reader's Digest to create an impact, while a minimum of four insertions will be required in the case of film fare.

Based on the available data, formulate (do not solve) an advertisement strategy for the company to maximize the expected effective exposure.

25. The XYZ Company is preparing a proposal for an advertising campaign for a client who is a manufacturer of law books. An optimal allocation of advertising funds to maximize the total number of exposures has to be made for the client. The characteristics of the three alternative publications are shown in the following table:

	Home beautiful (Rs)	Home and garden (Rs)	Care (Rs)
Cost/advertisement	600	800	450
Max. number of ads	12	24	12
Min. number of ads	3	6	2
Characteristics:			
Homeowner	80%	70%	20%
Income: Rs 10,000 or more	70%	80%	60%
Occupation: gardener	15%	20%	40%
Audience size	6,00,000	80,000	3,30,000

The relative importance of the three characteristics is: Homeowner, 0.4;Income,0.2;Gardener, 0.4. The advertising budget is Rs 20,000. Formulate the LP problem to find the most effective number of exposures in each magazine.

Problems on Finance:

26. An investor has money making activities A_1, A_2, A_3 and A_4. He has only one lakh rupee to invest. In order to avoid excessive investment, no more than 50% of the total investment can be placed in activity A_2 and/or activity A_3. Activity A_1 is very conservative, while activity A_4 is speculative. To avoid excessive speculation, at least Re 1must be invested in activity A_1 for every Rs 3 invested in activity A_4. The data onthe return on investment are as follows:

Activity	Anticipated return on investment (%)
A_1	10
A_2	12
A_3	14
A_4	16

The investor wishes to known how much to invest in each activity to maximize thetotal return on the investment.

27. The board of directors of a company has given approval for the construction of a new plant. The plant will require an investment of Rs 50lakhs. The required funds will come from sale of a proposed bond issue and loans from two financial corporations. For the company, it will not be possible to sell more than Rs 20 lakh in bond at the proposed rate is of 12%. Financial corporation A will loan upto Rs 30 lakh at an interest rate of 16% but insist that the amount of bond debt plus the amount owned to financial corporation B be no more than twice the amount owned to financial corporation A. financial corporation B will loan an amount upto the amount loaned byfinancial corporation A but at an interest rate of 18%. Formulate the LP model to determine the amount of funds to obtain from each source in a manner that minimizes total annual interest charges.

28. An investor wishes to diversify his portfolio and makes due allowance for a long term potentialities, but at the same time maximize his current dividend income. He has considered the various securities in which he might invest, and has classified them into four types:
Type A: Relatively high element of risk, with commensurately high dividend and

considerable growth potential.

Type B: speculative stocks with considerable risk, high dividends, but less growth potential than type A.

Type C: Stocks with little risk, considerable growth potential, but relatively low dividend income at present.

Type D: Stock with little risk, not much growth potential and fairly high dividends. Because of the element of risk, the investor wishes to restrict purchases of types Aand B of not more than 30% of his investment.

To enhance prospects for long term growth of his investments, he wishes to have at least 40% of his total outlay in types A and C. Within these restrictions, he wishes to maximize his current dividend income. Total investment is Rs 1, 00,000. Dividend returns on the four types of investments are A:6%, B:7%, C:3%, D:5%.

Formulate this problem as an LP model to suggest the total investment to be allocated.

29. The Agro Promotion Bank is trying to select an investment portfolio for a cotton farmer. The bank has chosen a set of five investment alternatives, with subjective estimates of rates of return and risk as follows:

Investment	Annual rate of return	Risk
Tax-free municipal bonds	6.0	1.3
Corporate bonds	8.0	1.5
High grade common stock	5.0	1.9
Mutual fund	7.0	1.7
Real estate	15.0	2.7

The bank officer in charge of the portfolio would like to maximize the average annual rate of return on the portfolio. However, the wealthy investor has specified that the average risk of the portfolio should not exceed 2.0 and does not want more than 20% of the investment to be put into real estate. Formulate this problem as an LP model.

Miscellaneous Problems:

30. A trucking firm has received an order to move 3,000 tonnes of industrial material to a destination 1,000km away. The firm has available at the moment a fleet of 150 class A 15-tonne trailer trucks and another fleet of 100 class B-ton trailer trucks. The operating costs of these trucks are Rs3 and Rs 4 per tonne per km respectively. Based on past experience the firm has a policy of retaining at least one class A truck with every two class B trucks in reserve. It is desired to know how many of these two classes of vehicles should be despatched to move the material at minimal operating costs. Formulate the above problem as a LPP.

31. A ship has three cargo loads-forward, after and centre. The capacity limits are:

	Weight (kg)	Volume (cu.cm)
Forward	2000	1,00,000
Centre	3000	1,35,000
After	1500	30,000

The following cargos are offered. The ship owner may accept all or any part of each commodity:

Commodity	Weight (kg)	Volume (cu.cm)	Profit inRs per kg
A	6000	60	60
B	4000	50	80
C	2000	25	50

In order to preserve the trim of the ship, the weight in each must be proportional to the capacity in kg. The cargo is to be distributed so as to maximize profit. Formulate this problem as an LP model.

32. Raj, a retired government officer, has recently received his retirement benefits, viz.., provident fund, gratuity etc. He is contemplating how much money he should invest in various alternatives open to him so as to maximize return on investment. The investment alternatives are: government securities, fixed deposit of a public limited company, equity shares, time deposit in a bank, and house construction. He has made a subjective estimate of the risk involved on a five-point scale. The data on the return on investment, the number of years for which the funds will be blocked to earn this return on investment and the subjective risk involved are as follows:

	Return (%)	Number of years	Risk
Government securities	6	15	1
Company deposits	13	3	3
Time deposits	10	5	2
Equity share	20	6	5
House construction	25	10	1

He was wondering what percentage of funds he should invest in each alternative so as to maximize the return on investment. He decided that the risk should not be more than 4, and funds should not be locked up for more than 15years. He would necessarily invest at least 25% in house construction.

33. A dealer in used scooters wishes to stock up his lot to maximize his profit. H can select scooters A, B and C which are valued on whole sale al Rs 5,000, Rs 7,000 and Rs 3,000 respectively. These can be sold at Rs 6,000, Rs 8500 and Rs 10,500 respectively. For each type of scooter, the probabilities of sale sold at Rs 6,000, Rs 8500 and Rs 10,500 respectively. For each type of scooter, the probabilities of sale sold at Rs 6,000, Rs 8500 and Rs 10,500 respectively. For each type of scooter, the probabilities of sale are:

Type of scooter:	A	B	C
Prob. Of sale in 90days:	0.7	0.8	0.6

For every two scooter of B-type he should buy one scooter of type A or C. If he has Rs 1,00,000 to invest, what should he buy to maximize his expected gain. Formulate this problem as an LP model.

34. Ex- servicemen Airport Services Company is considering the purchase of new vehicles for the transportation between the Delhi Airport and hotels in the city. There are three vehicles under consideration-station wagons, mini buses and large buses. The purchase price would be Rs 1, 45,000 for each station wagon, Rs 2, 50,000 for a mini bus and Rs 4, 00,000 for a large bus. The board of directors has authorized a maximum amount of Rs 50, 00,000 for these purchases. Because of the heavy air travel the new vehicles would be utilized at maximum capacity regardless of the type of vehicles purchased .The expected net annual profit would be Rs 15,000 for the station wagon, Rs 35,000 for the mini bus, and Rs 45,000 for the large bus. The company has hired 30 new drivers for the new vehicles. They are qualified drivers for all the three types of vehicles. The maintenance department has the capacity to handle an additional 80 station wagons. A mini bus is equivalent to 18pt station wagons and each large bus is equivalent to two station wagons in terms of their use of the maintenance department. Determine the optimal number of each type of vehicle to be purchased in order to maximize profit.

35. A manufacturer produces three models (I,II and III) of a certain product. He uses two types of raw material – A and B – of which 4000 and 6000 units are available respectively. The raw material requirements per unit of the three models are given below:

Raw Material	Requirements per unit of given model		
	I	II	III
A	2	3	5
B	4	2	7

The labour time for each unit of model I is twice that of model II and three times that of model III. The entire labour force of the factory can produce the equivalent of 2500 units of model I.A market survey indicates that the minimum demand for the three models is 500,500 and 375 units respectively. However, the ratios of the number of units produced must be equal to 3:2:5.Assume that the profit per unit of model I, II and III is Rs 60, Rs 40 and Rs 100 respectively. Formulate the problem as a linear programming model to determine the number of units of each product that will maximize profit.
Let x_1, x_2, x_3 = number of units of type I, II & III model respectively.

36. A company manufactures two models of garden roller—X and Y. when preparing the 1990 budget it was found that the limitations on capacity were represented by the following weekly production maxima:

Model	Foundry	Machine-shop	Contribution per model
X	160	200	Rs 120
Y	240	150	Rs 90

In addition the material required for model X was in short supply and only sufficient for 140 units per week could be guaranteed for the year. Formulate this problem as an LP model to determines the optimal combination of output.

37. The managers of several cattle feed lots of interested in determining how many of each of several types of livestock feeds to purchase in order to satisfy the nutritional requirements for their livestock. They wish to purchase such food in a manner that minimizes the cost of feeding their livestock. Relevant cost and nutritional data are as below:

Required nutrient	Units of nutritional element				Minimum nutrient requirements
	Alfa	Corn	Soyabeans	Sorghum	
A	40	50	30	60	500
B	30	60	35	40	750
C	25	30	25	50	600
Cost per unit (Rs)	1.00	1.25	0.95	1.35	

Formulate this problem as the LP model.

38. Old hens can be brought at Rs 2 each and young ones at Rs 5 each. The old hens lay 3 eggs per week and the young ones 5 eggs per week, each egg being worth 30 paise. A hen costs Re 1 per week to feed. I have only Rs 80 to spend for hens, hoe many of each kind should I buy to give a profit of more than Rs. 6 per week, assuming that I cannot house more than 20 hens. Formulate this problem as an LP model and solve graphically.

39. A pension fund manager is considering investing in two shares A and B. It isestimated that:
    i. Share A will earn can be dividend of 12%per annum and share B,4% per annum.
    ii. Growth in the market value in one year of share A will be 10paise per Re 1

invested and in B, 40 Paise per Re 1 invested.

He requires to invest the maximum total sum which will give:

i. Dividend income of at least Rs 600 per annum; and

ii. Growth in one year of at least Rs 1,000 on the initial investment.You are required to:

a. State the mathematical formulation of the problem.

b. Compute the minimum sum to be invested to meet the manager's objective byusing the graphic method.

40. A scrap metal dealer has received an order from a customer for at least 2,000 kg of scrap metal. The customer requires that at least 1,000 kg of the shipment of the metalbe high quality copper that can be melted down and used to produce copper tubings. Furthermore, the customer will not accept delivery of the order if it contains morethan 175 kg of metal that he deems unfit for commercial use, i.e. metal that containsan excessive amount of impurity and cannot be melted down and refined profitability. The dealer can purchase scrap metal from two different suppliers in unlimitedquantities with the following percentage (by weight) of high quality copper and unfitscrap

	Supplier	Supplier
Copper	25%	75%
Unfit scrap	5%	10%

The cost per kg of metal purchased from supplier A and supplier B is Re 1 and Rs 4 respectively. The problem is to determine the optimal quantities of metal for the dealer to purchase from each of the two suppliers.

Graphical Method:

The graphical solution procedure

i. Consider each inequality constraint as equation.

ii. Plot each equation on the graph as each one will geometrically represent a straight line.

iii. Shade the feasible region. Every point on the line will satisfy the equation of the line. If the inequality constraint corresponding to that line is '≤' then the region below the line lying in the first quadrant is shaded. Similarly for '≥' the region above the line is shaded. The points lying in the common region will satisfy the constraints. This common region is called *feasible region*.

iv. Choose the convenient value of Z and plot the objective function line.

v. Pull the objective function line until the extreme points of feasible region.

vi. In the maximization case this line will stop far from the origin and passing through at least one corner of the feasible region.

vii. In the minimization case, this line will stop near to the origin and passing through at least one corner of the feasible region.

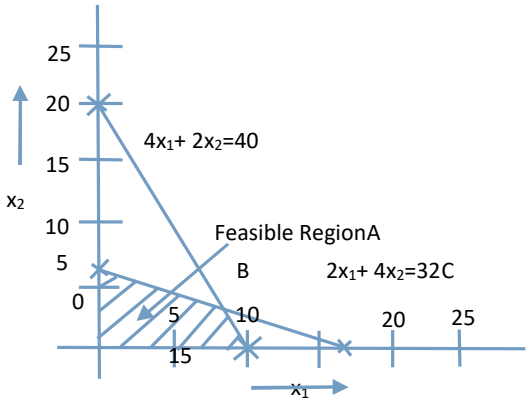
viii. Read the co-ordinates of the extreme points selected in step 5 and find the maximum or minimum value of Z.

ExtremePoints	Co-ordinates	Objective Function	Value
A	(x_1,x_2)		
B	(x_1,x_2)		
C	(x_1,x_2)		

Problems on Graphical Method:

43. Solve the following Linear Programming Problem by Graphical Method:Max Z = $80x_1 + 55x_2$
Subject to
$4x_1 + 2x_2 \leq 40$
$2x_1 + 4x_2 \leq 32 x_1 \geq 0 , x_2 \geq 0$

Solution: The first constraint $4x_1+2x_2 \leq 40$, written in a form of equation$4x_1 + 2x_2 = 40$
Put $x_1 = 0$, then $x_2 = 20$Put $x_2 = 0$, then $x_1 = 10$
The coordinates are (0,20) and (10,0)
The second constraint $2x_1+4x_2 \leq 32$, written in a form of equation$2x_1 + 4x_2 = 32$
Put $x_1 = 0$, then $x_2 = 8$ Put $x_2 = 0$, then $x_1 = 16$
The coordinates are (0,8) and (16,0)The graphical representation is



The corner points of feasible region are A, B and C. So the coordinates for the corner pointsare A (0,8); B (8,4); C (10,0)

Extreme Points	Co-ordinates	Objective Function $Z_{max} = 80x_1 + 55x_2$	Value
A	(0,8)	80(0)+55(8)	440
B	(8,4)	80(8)+55(4)	860
C	(10,0)	80(10)+55(0)	800

The maximum value is obtained at the point B.

Therefore Max Z = 860 and $x_1 = 8$, $x_2 = 4$

44. Solve the following Linear Programming Problem by Graphical Method:Max Z = $2x_1 + 10x_2$

Subject to

$2x_1 + 5x_2 \leq 16$

$6x_1 + 0x_2 \leq 30 x_1 \geq 0$, $x_2 \geq 0$

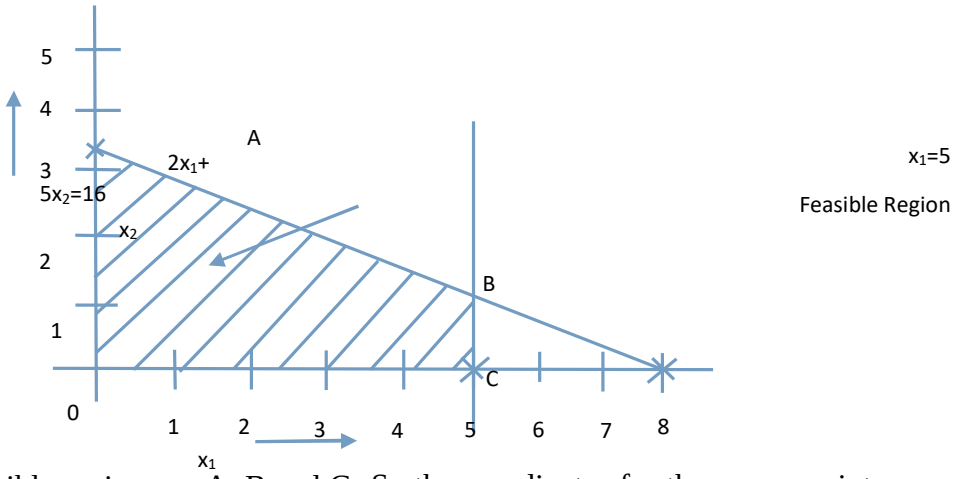
Solution:

The first constraint $2x_1+5x_2 \leq 16$, written in a form of equation $2x_1+5x_2 = 16$Put $x_1 = 0$, then $x_2 = 16/5 = 3.2$

Put $x_2 = 0$, then $x_1 = 8$

The coordinates are (0,3.2) and (8,0)

The second constraint $6x_1 + 0x_2 \leq 30$, written in a form of equation $6x_1 = 30 \rightarrow x_1 = 5$The graphical representation is



The corner points of feasible region are A, B and C. So the coordinates for the corner points are A (0,3.2); B (5,1.2); C (5, 0)

Extreme Points	Co-ordinates	Objective Function $Z_{max} = 2x_1 + 10x_2$	Value
A	(0,3.2)	2(0)+10(3.2)	32
B	(5,1.2)	2(5)+10(1.2)	22
C	(5, 0)	2(5)+10(0)	10

The maximum value is obtained at the point A. ThereforeMax Z = 32 and $x_1 = 0$, $x_2 = 3.2$

45. Solve the following Linear Programming Problem by Graphical Method:

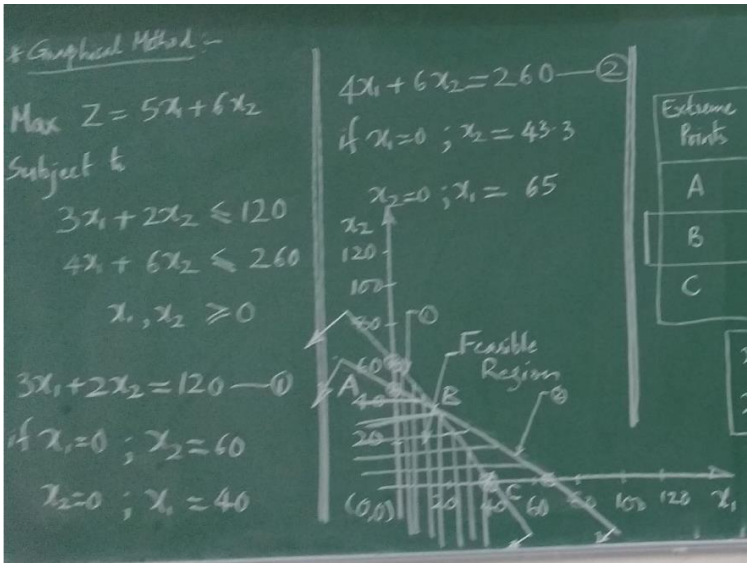
Maximize $Z = 5x_1 + 6x_2$

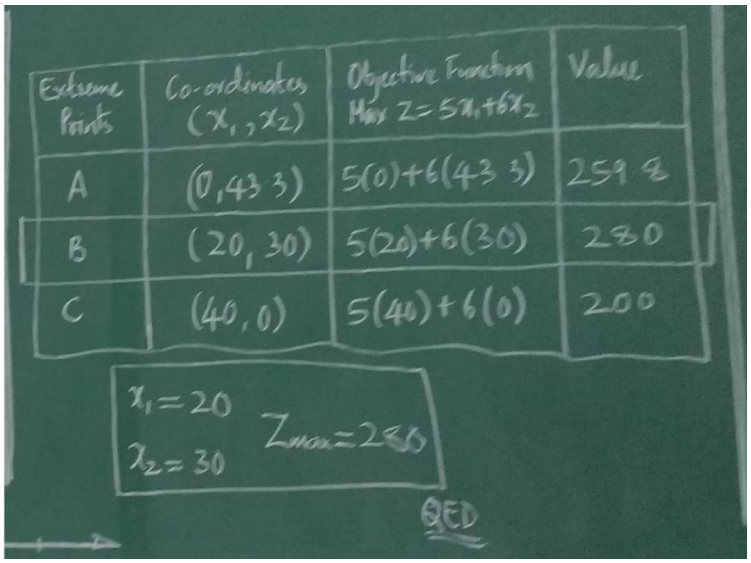
Subject to

$3x_1 + 2x_2 \leq 120$

$4x_1 + 6x_2 \leq 260$

x_1 and $x_2 \geq 0$





46. Solve the following Linear Programming Problem by Graphical Method:

Minimize $Z = 3x_1 + 6x_2$

Subject to

$4x_1 + x_2 \geq 20$ $x_1 + x_2 \leq 20$ $x_1 + x_2 \geq 10$ x_1 and $x_2 \geq 0$

47. Solve the following Linear Programming Problem by Graphical Method:

Minimize $Z = 600x_1 + 400x_2$

Subject to

$1500x_1 + 1500x_2 \geq 20000$

$3000x_1 + 1000x_2 \geq 40000$

$2000x_1 + 5000x_2 \geq 44000$

x_1 and $x_2 \geq 0$

48. Solve the following Linear Programming Problem by Graphical Method:

Maximize $Z = 40x_1 + 100x_2$

Subject to

$12x_1 + 6x_2 \leq 3000$

$4x_1 + 10x_2 \leq 2000$

$2x_1 + 3x_2 \leq 900$

x_1 and $x_2 \geq 0$

Special Cases in Graphical Method:

i. **Multiple Optimal Solution:**

49. Solve by using graphical methodMax Z = $4x_1 + 3x_2$

Subject to

$4x_1 + 3x_2 \leq 24 x_1 \leq 4.5$

$x_2 \leq 6$

$x_1 \geq 0$, $x_2 \geq 0$

Solution:

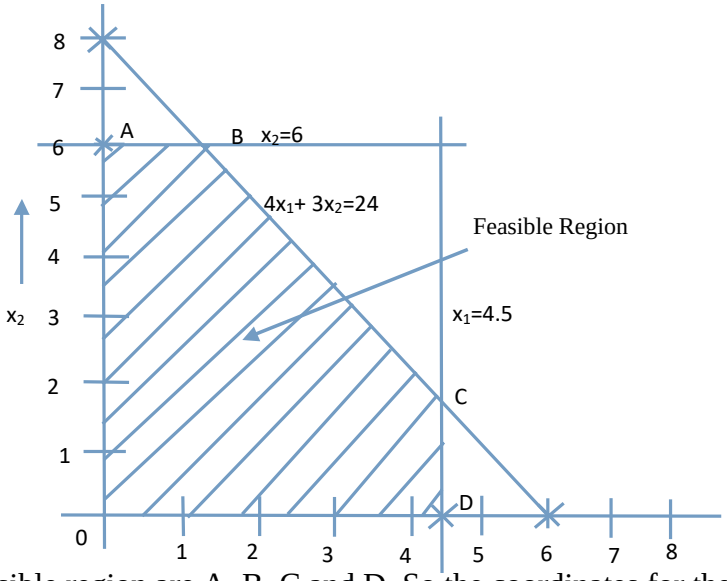
The first constraint 4x 1 + 3x 2 ≤ 24, written in a form of equation$4x_1 + 3x_2 = 24$

Put x_1 =0, then x_2 = 8Put x_2 =0, then x_1 = 6

The coordinates are (0, 8) and (6, 0)

The second constraint $x_1 \leq 4.5$, written in a form of equation$x_1 = 4.5$

The third constraint $x_2 \leq 6$, written in a form of equation$x_2 = 6$



The corner points of feasible region are A, B, C and D. So the coordinates for the cornerpoints are

A (0,6)

B (1.5,6) (Solve the two equations $4x_1 + 3x_2 = 24$ and $x_2 = 6$ to get the coordinates)

C (4.5, 2) (Solve the two equations $4x_1 + 3x_2 = 24$ and $x_1 = 4.5$ to get the coordinates)

D (4.5,0)

We know that Max Z = $4x_1 + 3x_2$

At A (0,6), Z = 4(0) + 3(6) = 18

At B (1.5,6), Z = 4(1.5) + 3(6) = 24

At C (4.5,2), Z = 4(4.5) + 3(2) = 24

At D (4.5,0), Z = 4(4.5) + 3(0) = 18

Max Z = 24, which is achieved at both B and C corner points. It can be achieved not only at B and C but every point between B and C. Hence the given problem has multiple optimal solutions.

ii. **No Optimal Solution:**

50. Solve graphically

Max Z = $3x_1 + 2x_2$Subject to

$x_1 + x_2 \leq 1 x_1 + x_2 \geq 3$

$x_1 \geq 0$, $x_2 \geq 0$

Solution:

The first constraint $x_1 + x_2 \leq 1$, written in a form of equation $x_1 + x_2 = 1$Put x 1 =0, then $x_2 = 1$

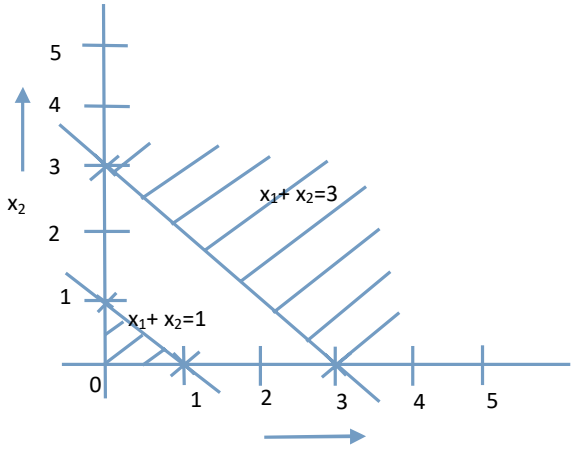
Put x 2 =0, then $x_1 = 1$

The coordinates are (0,1) and (1,0)

The first constraint $x_1 + x_2 \geq 3$, written in a form of equation $x_1 + x_2 = 3$Put x_1 =0, then $x_2 = 3$

Put x_2 =0, then $x_1 = 3$

The coordinates are (0,3) and (3,0)



x1

Unbounded Solution:

51. Solve by graphical methodMax Z = $3x_1 + 5x_2$ Subject to

$2x_1 + x_2 \geq 7$

$x_1 + x_2 \geq 6$

$x_1 + 3x_2 \geq 9$

$x_1 \geq 0$, $x_2 \geq 0$

Solution:

The first constraint $2x_1 + x_2 \geq 7$, written in a form of equation $2x_1 + x_2 = 7$Put $x_1 = 0$, then $x_2 = 7$

Put $x_2 = 0$, then $x_1 = 3.5$

The coordinates are (0,7) and (3.5,0)

The second constraint $x_1 + x_2 \geq 6$, written in a form of equation $x_1 + x_2 = 6$Put $x_1 = 0$, then $x_2 = 6$

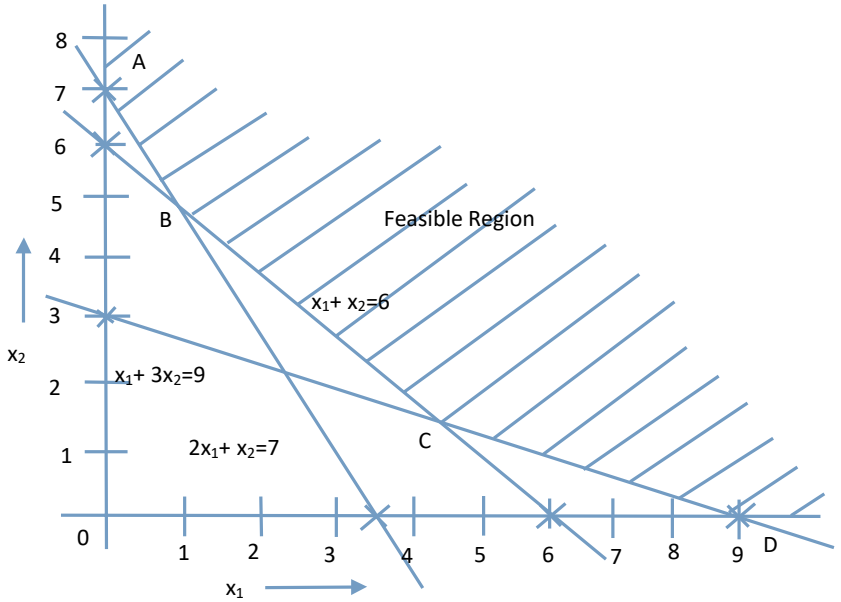
Put x2 =0, then $x_1 = 6$

The coordinates are (0,6) and (6,0)

The third constraint $x_1 + 3x_2 \geq 9$, written in a form of equation $x_1 + 3x_2 = 9$Put $x_1 = 0$, then $x_2 = 3$

Put $x_2 = 0$, then $x_1 = 9$

The coordinates are (0,3) and (9,0)



The corner points of feasible region are A, B, C and D. So the coordinates for the cornerpoints are

A (0,7)

B (1,5) (Solve the two equations $2x_1 + x_2 = 7$ and $x_1 + x_2 = 6$ to get the coordinates)

C (4.5,1.5) (Solve the two equations $x_1 + x_2 = 6$ and $x_1 + 3x_2 = 9$ to get the coordinates)D (9,0)

We know that Max Z = $3x_1 + 5x_2$

At A (0,7); Z = 3(0) + 5(7) = 35

At B (1,5); Z = 3(1) + 5(5) = 28

At C (4.5,1.5); Z = 3(4.5) + 5(1.5) = 21

At D (9,0); Z = 3(9) + 5(0) = 27

The values of objective function at corner points are 35, 28, 21 and 27. But there exists infinite number of points in the feasible region which is unbounded. The value of objective function will be more than the value of these four corner points i.e. the maximum value of the objective function occurs at a point at ∞. Hence the given problem has unbounded solution.